Main.java

It just contains main function itself. Acting like a core for starting the program.

What it does is:

1. **Gets user input** (Scanner class)
2. **Splits the user input into array**
3. **Checks input length**: Whether the user is intended to use simpleFraction() or complexFraction(). If none returns error
4. Regardless of the input length based on the indexing **converts every item of the array into integer** using wrapper function called **valueOf()**
5. As **valueOf()** may throw **NumberFormatException** all the logic is wrapped around try/catch block

FractionUntil.java

This is the code where the main logic is defined. Contains 2 special functions:

1. **simpleFraction():** From user input gets numerators and denominators. And then finds their non-reduced sum. For simplification there is a private static method called **simplify_and_print()**. This function takes the non-abbreviated sum and divides it with greatest common divisor of them (using user-defined **ebob()** function).
2. **complexFraction()**: Uses the same logic as simpleFraction() method. Only difference is this algorithm also takes signs as parameters, then uses them to decide what the user is intended to do with the input.

Both of these methods do not make any return. Main purpose behind that the description of the assignment 1 do not specifically written return but output. That is the reason why there is no any returns.

Usage

User needs to compile the files, and both of the java files have to be in the same directory. For compilation **javac** and for running the program **java** command would be sufficient. User input should be "**1, 2, 3, 4**" ("**-, 2, 4, +, 5, 6**") or likewise. Any unnecessary signs, commas, or characters will cause the program to throw an error (or run improperly) so they should be avoided.

Conclusion

With the connection of **Main.java** and **FractionUntil.java** files I have successfully managed to implements a basic yet error-free (I believe) code chunk. As the task intended all of these functions are static so that they can be called from any file without the need of new class object. The helper functions, however; are private so that they can only be used by the FractionUntil.java file and its methods.